Lecture Notes: Introduction to Astrophysics and Cosmology

Based on lectures by **Dr. Shmuel Bialy** in 2025-26
Notes created by Daniel Haim Breger

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These notes were typed in the 2026 winter semester *Introduction to Astrophysics and Cosmology* course taught by **Dr. Shmuel Bialy** at the Technion.

The lectures were given in Hebrew and were live-translated by me to English. The document is provided as is and likely contains many errors.

If you find any mistakes or typos please let me know at `danielbreger@campus.technion.ac.il`.

Contents

Lecture 1.

Thu, October 30, 2025

1 introduction

Since astrophysics is a very wide subject we won't be able to dive deep on each subject. The course is introductory so we'll be skipping some things like full solutions to integrals and the such.

1.1 So what is astrophysics?

Astrophysics and Cosmology could be broken down into:

Physics of stars

Solar systems planets, comets…

ISM Inter-Stellar Medium, Inter-Galactic Medium

Cosmology development, the big bang, dark energy/matter

Exotic objects black holes, neutron stars, white dwarfs…

Relativity gravitational waves, gravitational lensing

Galaxies types, development, active galactic nuclei, quasars

Explosions novas, supernovas, gamma ray bursts

and the methods used to research those things include

Observations
- **Telescopes** IR, radio, xray, gamma rays…
- **Gravitational waves**
- **Neutrinos**
- **Other particles (via probes)**

Labs

Computer simulations hydrodynamics, N-body simulations…

Analytical theory

We'll focus mostly on the analytical theory, but we'll mention the others too.

Lecture 2.

Mon, November 3rd, 2025

1.2 Regular (Baryonic) Matter

In our normal day-to-day life we meet protons, neutrons, and electrons. There are other particles we come across, such as neutrinos, muons, taus, etc. Another particle we meet often is the photon, and some others we won't get into.

When the universe came into being in the big bang quarks and photons were created. Those quarks then combined into protons and neutrons. Protons are basically hydrogen nuclei, so hydrogen was created in the big bang. For a short period of time after the big bang there were conditions for nuclear fusion across the early universe, so about 10% of the matter created from the big bang ended up being helium, and in terms of mass it was about 25% (Because Helium is heavier than hydrogen). Elements heavier than helium weren't created at any significant quantities in the big bang. If we plot this on a graph:

The portion of mass of hydrogen of the total mass is denoted by X, hydrogen by Y, and metals, which are all the elements heavier than helium and are denoted by $Z = \frac{M_{metals}}{M}$. For our sun:

$$X_\odot = 0.7381 \tag{2.1}$$
$$Y_\odot = 0.2485 \tag{2.2}$$
$$Z_\odot = 0.0134 \tag{2.3}$$

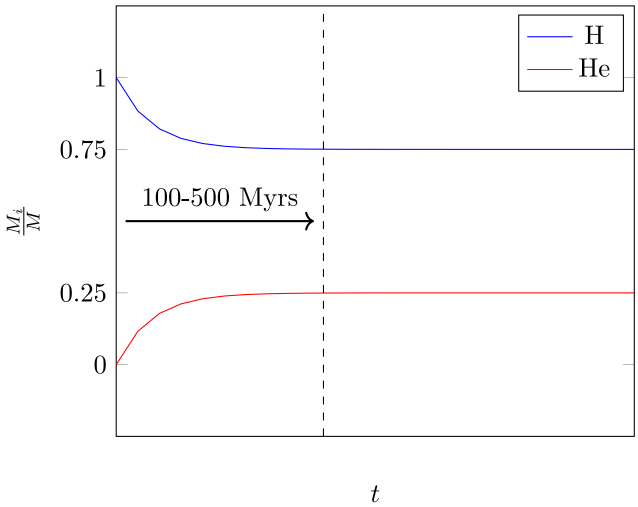


Figure 2.1: The portion of mass of the universe by element in the universe's early times.

The above was a discussion for gas and plasma. There is another kind of matter in space called dust. Dust is similar to the dust we know. It's particles about a micrometer in diameter made of water, carbon, silica, etc. Why do we mention this? In terms of the mass of the metals in the universe, about half of it is dust. Moreover, dust is a significant factor in observations of the universe, either in interfering in our ability to see things or in creating conditions that allow us to infer information. Dust also aids the creation of molecules by speeding up processes which are very slow in gasses.

An important definition for us will be the 'average mass' which we'll use in many formulas. Assume we have a box filled with hydrogen and some helium and some metals, all with the relative quantities that are in the sun. What would the average mass per particle be? Approximately a proton mass. And if the gas in the box is ionized? 0.5 proton masses. This is because we doubled the number of particles by freeing the electrons, which have approximately 0 mass relatively to the protons. This was a back of the napkin approximation just for sanity check and to have an idea of what we're expecting. The exact calculations:

$$\overline{m} = \frac{\sum\limits_j N_j m_j}{\sum\limits_j N_j} \tag{2.4}$$

Where j is each element type, N is their number, and m their mass. Rewriting in terms of X, Y, Z:

$$\frac{1}{\overline{m}} = \frac{\sum N_j}{\sum N_j m_j} \tag{2.5}$$
$$= \frac{M_j}{M_{tot}} \cdot \sum \frac{N_j}{M_j} \tag{2.6}$$
$$= \frac{1}{m_H} \left(X + \frac{1}{4} Y + \frac{1}{A_Z} Z \right) \tag{2.7}$$

where

$$M_j = \begin{cases} N_H m_H & H \\ N_{He} \cdot 4 m_H & He \\ N_Z A_Z & else \end{cases} \tag{2.8}$$

Sometimes the following definition is used:

$$\mu = \frac{\overline{m}}{m_H} \tag{2.9}$$

and if we plug in the numbers for the sun we indeed find that it's very close to 1. We won't do this explicitly because there's a correction for the formula for a non-neutral gas, which our sun is not. For a completely ionized gas, i.e all the electrons are free, we just need to change N_j:

$$N_j = \begin{cases} 1 \to 2 & H \\ 1 \to 3 & He \\ 1 \to \left\langle \frac{A_Z}{2} \right\rangle & Else \end{cases} \tag{2.10}$$

Which gives us, for a completely ionized gas:

$$\mu = \left[2X + \frac{3}{4} Y + \frac{1}{2} Z \right]^{-1} \tag{2.11}$$

and if you insert the numbers for our sun we find

$$\langle A \rangle = 15 - 16 \tag{2.12}$$
$$\mu_{\odot} \simeq 0.6 \tag{2.13}$$

Lecture 3.

Thu, November 6th, 2025

1.3 The meaning of the symbol $\simeq$

The symbol $\simeq$ usually implies a good understanding of the physics behind the calculations. It says there's a good understanding of what can be considered negligible and what can't.

The luminosity of the sun follows

$$L_\odot = \dot{E}_{\text{nuc}} + |\dot{E}_G| \tag{3.1}$$

Observationally we can measure the luminosity and find

$$L_\odot = 3.8 \times 10^{33}\,\text{erg}\,\text{s}^{-1} \tag{3.2}$$

and we can say that

$$L_\odot \simeq \dot{E}_{\text{nuc}} \tag{3.3}$$

Let's look at the total energy the Sun will output in its entire life E_{tot}.

$$\begin{aligned} E_{tot} &= L_\odot \cdot t_\odot && (3.4)\\ &\simeq (3.8 \times 10^{33}\,\text{erg}\,\text{s}^{-1})(4.6\,\text{Gyr}) && (3.5)\\ &= 5.6 \times 10^{50}\,\text{erg} && (3.6) \end{aligned}$$

Let's look at what contributes more to the energy of the sun, the gravitational or the nuclear energy.

$$\begin{aligned} |E_G| &= \int \frac{GM(r)}{r}\,\mathrm{d}m && (3.7)\\ &= \int 4\pi\rho GM(r)r\,\mathrm{d}r && (3.8)\\ (\text{assuming } \rho = const.) &= \frac{3}{5}G\frac{M_\odot^2}{R_\odot} && (3.9)\\ &= 2 \times 10^{48}\,\text{erg} && (3.10) \end{aligned}$$

To estimate the nuclear energy the sun produces we can look at the nuclear fusion process in which $4H \to He$ which gives a mass difference of

$$\begin{aligned} \Delta m &= M(4H) - M(He) && (3.11)\\ \to \Delta E &= \Delta mc^2 && (3.12) \end{aligned}$$

defining

$$\eta = \frac{\Delta E}{M_{He}c^2} = 0.7\% \tag{3.13}$$

then the nuclear energy produced in the sun is

$$\eta \cdot M_\odot c^2 \tag{3.14}$$

and using this we can find the age of the sun

$$t_\odot = \frac{E_{nuc}}{L_\odot} \approx 100\,\text{Gyr} \tag{3.15}$$

but since the sun hasn't burned through all the hydrogen it has access to, we can replace

$$M_\odot \to M_{core} \approx 5\,\% M_\odot \tag{3.16}$$

which would give us a better estimate.

2 The Structure of Stars

We'll use the Sun as an example, then we'll hit equations.

2.1 The Sun

Our goal in the coming weeks will be to derive differential equations to figure out how the different profiles behave. As for some numbers:

$$\begin{aligned} R_\odot &= 7\times 10^{10}\,\text{cm} & T_C &= 1.5\times 10^{7}\,\text{K} \\ \rho_C &= 150\,\text{g}\,\text{cm}^{-3} & \overline{\rho} &= 1.4\,\text{g}\,\text{cm}^{-3} \\ P_C &= 10^{11}\,\text{Atm} & M_\odot &= 2\times 10^{33}\,\text{g} \\ L_\odot &= 3.8\times 10^{33} erg/s & & \end{aligned}$$

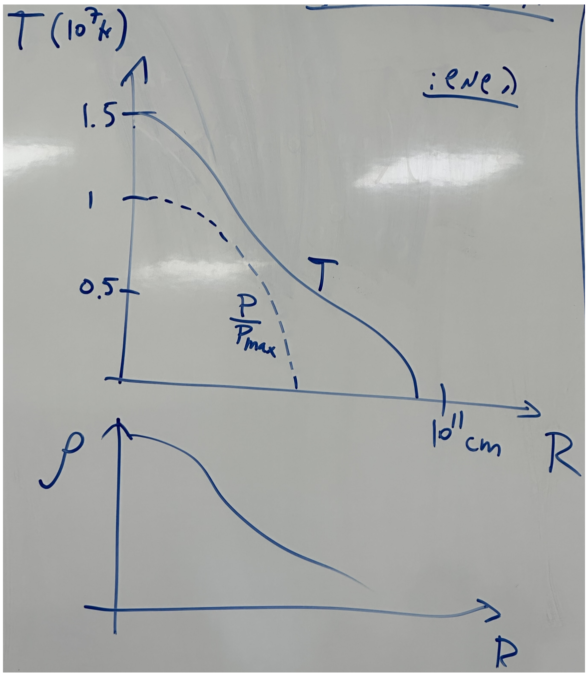


Figure 3.1: The temperature in the sun as a function of the distance from it core, the relative pressure compared to the core, and the density.

2.2 Star Structure Equations

The equations describing the structure of stars are:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -G\frac{M_r \rho}{r^2} \tag{3.17}$$

$$\frac{\mathrm{d}M_r}{\mathrm{d}r} = 4\pi r^2 \rho \tag{3.18}$$

$$\frac{\mathrm{d}L_r}{\mathrm{d}r} = 4\pi r^2 \rho \epsilon \tag{3.19}$$

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3\kappa\rho}{4acT^3}\frac{L_r}{4\pi r^2} \tag{3.20}$$

The first equation is called the hydrostatic equation and is equivalent to conservation of momentum, the second is the continuum equation and is equivalent to conservation of mass, the third is the energy production equation and is equivalent to conservation of energy, and the final one is the radiation diffusion equation.

2.3 The Hydrostatic Equation

The equation can be derived from an equality of forces. We look at a shell at radius r with mass $\mathrm{d}m$ and ask which forces act on it. On one hand there's gravitation pulling it inwards and on the other side there's a force resulting from a difference in pressure.

$$|\mathrm{d}F_g| = \frac{GM_r\,\mathrm{d}m}{r^2} \tag{3.21}$$
$$= \frac{GM_r}{r^2}4\pi r^2\rho\,\mathrm{d}r \tag{3.22}$$
$$|\mathrm{d}F_P| = A\cdot(P(r) - P(r+\mathrm{d}r)) \tag{3.23}$$
$$= 4\pi r^2\left(-\frac{\mathrm{d}P}{\mathrm{d}r}\,\mathrm{d}r\right) \tag{3.24}$$

equating the two forces we can find

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho\frac{GM_r}{r^2} \tag{3.25}$$

which is indeed the hydrostatic equation.

Applying The Equation To Earth

As a back of the envelope calculation, we can use the equation to find the gravitational acceleration of earth. Noticing that

$$\frac{GM_r}{r^2} = g \tag{3.26}$$

we can rewrite the hydrostatic equation as

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho g \tag{3.27}$$

We can also find the speed of sound

$$c_s = \sqrt{\gamma\frac{\mathrm{d}P}{\mathrm{d}\rho}} \tag{3.28}$$

which means

$$\frac{\mathrm{d}P}{\mathrm{d}\rho} \approx c_s^2 \tag{3.29}$$

so

$$g \sim \frac{\mathrm{d}P}{\mathrm{d}\rho}\frac{1}{h} \tag{3.30}$$
$$= \frac{(340\,\mathrm{m\,s^{-1}})^2}{10\,\mathrm{km}} \tag{3.31}$$
$$= 10\,\mathrm{m/s^2} \tag{3.32}$$

Lecture 4.

Mon, November 10th, 2025

2.4 The State Equation

The gas in stars is to a good approximation an ideal gas so we can use the state equation of an ideal gas

$$P = Nk_BT \tag{4.1}$$

we can multiply and divide by the average mass of each particle

$$P = \frac{n\overline{m}}{\overline{m}}k_BT = \frac{\rho}{m_p\mu}k_BT \tag{4.2}$$

A quick aside on why the state equation is true

This is taken from Feynman's lectures on physics volume 2 chapter 39-2. Assume we have a wall of area A and particles with density n, each having velocity v_x and they impact the wall head on. As we know $F = \frac{\mathrm{d}p}{\mathrm{d}t}$.

The distance the particles travel over a period of time of δt is $\delta x = v_x\delta t$. So the rate at which particles hit over the same time period is $\delta N = \frac{1}{2}nAv\delta t$. Each particles transfers a momentum of $2p_1 = 2mv$ therefore the total momentum transferred is

$$\delta p = mnAv_x^2\delta t \tag{4.3}$$

therefore the force is

$$F = \frac{\mathrm{d}p}{\mathrm{d}t} = mnAv_x^2 \tag{4.4}$$

and the pressure is

$$P = \frac{F}{A} = mnv_x^2 \tag{4.5}$$

and if the velocity isn't a constant but rather there's a distribution of velocities then

$$P = mn\langle v_x^2\rangle \tag{4.6}$$

when in 3 dimensions

$$\langle v^2\rangle = \langle v_x^2\rangle + \langle v_y^2\rangle + \langle v_z^2\rangle = 3\langle v_x^2\rangle \tag{4.7}$$

So the pressure is

$$P = \frac{1}{3}mn\langle v^2\rangle \tag{4.8}$$

If we define temperature as follows

$$\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_BT \tag{4.9}$$

and then insert this into our equation for pressure we get

$$P = nk_BT \tag{4.10}$$

Thermal energy is

$$U = \frac{1}{2}m\langle v^2\rangle N \tag{4.11}$$

so its density is

$$u = \frac{1}{2}m\langle v^2 c\rangle n \tag{4.12}$$

therefore we can insert this into the expression for pressure we have and find

$$P = \frac{2}{3}u \tag{4.13}$$

This is correct for mono-atomic gas such as hydrogen. In the more complicated case we can define the adiabatic index

$$\gamma_{ad} = \frac{C_p}{C_v} = 1 + \frac{2}{f} \tag{4.14}$$

where f is the number of degrees of freedom, and then we can find that

$$P = (\gamma_{ad} - 1)u \tag{4.15}$$

2.5 Generalizing To Radiation Pressure

The distribution of the velocities of the massive particles is the Boltzmann distribution. On the other hand, the distribution of the momenta of the photons is the Planck distribution:

$$\frac{\mathrm{d}n}{\mathrm{d}\nu} = n_\nu = \frac{8\pi\nu^2}{c^3}\frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \tag{4.16}$$

and

$$u_\nu = n_\nu h\nu = \frac{8\pi h\nu^3}{c^3}\frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \tag{4.17}$$

To get the pressure, rather than doing an entirely new development, we can say our previous development of the pressure using a wall and particles was good it just needs some adjustments. We had:

$$P = \frac{1}{3}mn\langle v^2\rangle = \langle\frac{1}{3}nv \cdot mv\rangle = \langle\frac{1}{3}nv \cdot p\rangle \tag{4.18}$$

inserting the momentum and velocity of photons we find

$$P = \langle\frac{1}{3}nc\frac{h\nu}{c} \tag{4.19}$$

$$= \langle\frac{1}{3}nh\nu\rangle \tag{4.20}$$

$$= \frac{1}{3}\langle u\rangle \tag{4.21}$$

$$= \frac{1}{3}u \tag{4.22}$$

And if we equate this to the more general pressure equation using the adiabatic constant we find

$$\gamma_{\mathrm{photons}} = \frac{4}{3} \tag{4.23}$$

to rewrite this in terms of temperature we remember that

$$u = \int_0^\infty u_\nu \, d\nu = \int \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \tag{4.24}$$

$$= [x = \frac{h\nu}{k_B T}] = 8\pi \frac{(kT)^4}{(hc)^3} \int_0^\infty x^3 \frac{dx}{e^x - 1} \tag{4.25}$$

$$= 8\pi \frac{(kT)^4}{(hc)^3} \cdot \frac{\pi^4}{15} \tag{4.26}$$

we can define a constant

$$a = \frac{8\pi^5 k_B^4}{15(hc)^3} \tag{4.27}$$

and have

$$u = aT^4 \tag{4.28}$$

and find that the total pressure is

$$P_{tot} = nk_B T + \frac{1}{3} aT^4 \tag{4.29}$$

2.6 Nuclear Fusion

quick look at the periodic table. quick look at the standard model. the dominant nuclear fusion process is the pp-chain.

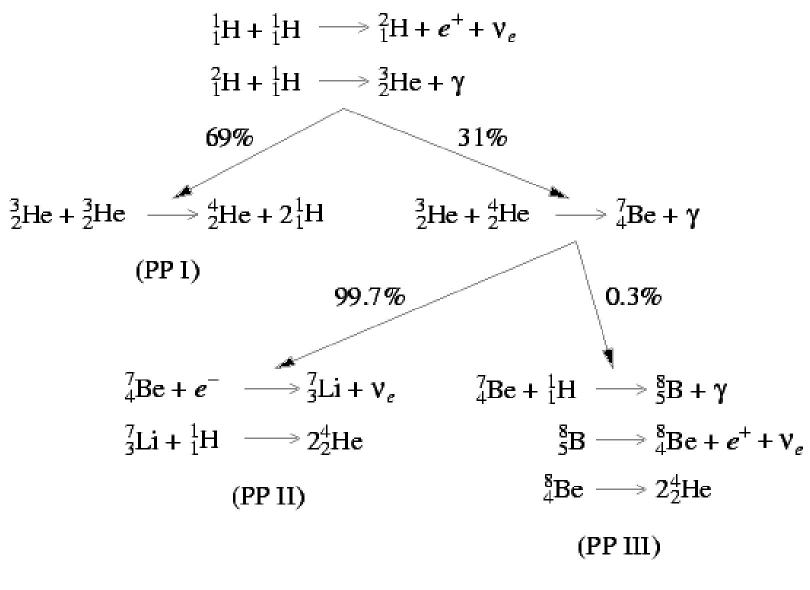


Figure 4.1: https://burro.cwru.edu/academics/Astr221/StarPhys/ppchain.html

The efficiency of the process is

$$\eta = \frac{E}{Mc^2} = \frac{26.7MeV}{4m_p c^2} = 0.7\% \tag{4.30}$$

for context of how much energy that is, we know that the sun's luminosity is $l_\odot = 4 \times 10^{33}$ erg/sec. This means that per second it produces 4×10^{26} J, which is 10^6 times the yearly consumption of all humanity.

Lecture 5.

Thu, November 13th, 2025

To find the luminosity we need to also figure out the pace of the processes, which we'll call r_{xi} whose units are $\mathrm{s}^{-1}\mathrm{cm}^{-3}$, x indicating the target particle being hit by the particle i. First though let's check if nuclear fusion can happen in the sun at all.

The electrostatic potential (in cgs) is given by

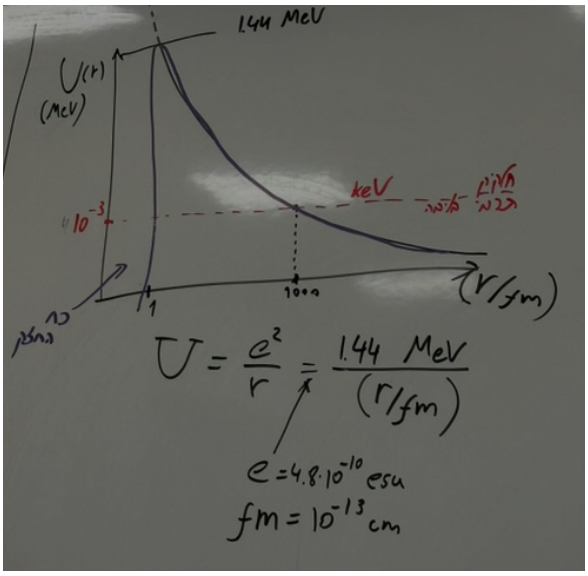
$$U = \frac{e^2}{r} = \frac{1.44\,\mathrm{MeV}}{(r/fm)} \tag{5.1}$$

where

$$e = 4.8 \times 10^{-10}\,\mathrm{esu} \qquad fm = 10^{-13}\,\mathrm{m} \tag{5.2}$$

and the temperature at the core of the sun is $T_c \simeq 1 \times 10^7\,\mathrm{K}$ and using the expression for the energy at the core

$$\frac{3}{2}k_B T = E \implies E_c \sim keV \tag{5.3}$$



This means that the thermal particles have only 0.1% of the energy required to jump the energy barrier imposed by the coulomb force's repulsion to reach the area the strong force attracts them. Seemingly if the system were purely classical, fusion could not happen. Let's consider an edge case where maybe fusion happens because a random small number of particles just happen to have a high enough energy to pass the barrier.

To do that we can calculate how many particles on average there are that have an energy with approximately a mega-electronvolt of energy. For a rough estimate we can use statistical mechanics

$$e^{-\frac{E}{k_B T}} \sim e^{-1000} = 10^{-435} \tag{5.4}$$

where we inserted the temperature at the core of the sun and the energy we want to have to breach the barrier. The number is so small that there would on average be less than one particle in the entire universe that would have that energy. For that reason we didn't calculate it accurately, since it wouldn't have mattered.

We'll need to take into consideration the effects of quantum tunneling, but before we do that, we need to discuss the action cross-section. We'll imagine a ball x with effective surface area σ being bombarded with particles i.

The number of particles hitting the area is given by

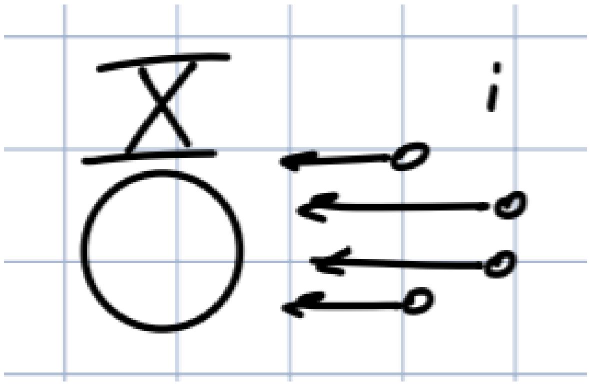
$$\mathrm{d}N = \sigma v \delta t \cdot n_i \tag{5.5}$$

so the rate at which they hit is

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \sigma v n_i \tag{5.6}$$

If we have an environment with many x particles then

$$r_{xi} = \sigma v n_i \cdot n_x \tag{5.7}$$



where n_x is the density of the particles x.
In the more general case if we have a distribution of velocities then we have

$$v \implies v(E) \qquad\qquad n_i \implies \frac{\mathrm{d}n_i}{\mathrm{d}E}\,\mathrm{d}E \tag{5.8}$$

and then

$$r_{xi} = \int \sigma(E) v(E) n_x \frac{\mathrm{d}n_i}{\mathrm{d}E}\,\mathrm{d}E \tag{5.9}$$

with the Stefan-Boltzmann law being

$$\frac{\mathrm{d}n_i}{\mathrm{d}E} = \frac{2n_i}{\sqrt{\pi}} \frac{\sqrt{E}}{(k_B T)^{3/2}} e^{-\frac{E}{k_B T}} \tag{5.10}$$

our goal now is to find $\sigma(E)$. For quantum tunneling we'll write

$$\sigma = \sigma_0 P \tag{5.11}$$

where P is the chance to tunnel and σ_0 is the effective area. As for σ_0:

$$\begin{aligned}
\sigma_0 &\approx \pi R^2 && (5.12)\\
&= \pi \lambda_{DB}^2 && (5.13)\\
&= \pi \left(\frac{h}{p}\right)^2 && (5.14)\\
&= \pi \frac{h^2}{2\mu E} && (5.15)
\end{aligned}$$

where μ is the reduced mass, since we're working with a collision of two particles. or, in a general more accurate form

$$\sigma_0 = \frac{S}{E} \tag{5.16}$$

To find the chance to tunnel we'll refer to quantum mechanics but this isn't a course on quantum mechanics so it's just a quick overview.

The Schrödinger equation is

$$\frac{\hbar^2}{2\mu}\nabla^2\Psi = (V(r) - E)\,\Psi \tag{5.17}$$

where our energy is

$$E = \frac{3}{2}k_BT = \frac{z_1z_2e^2}{r_1} \tag{5.18}$$

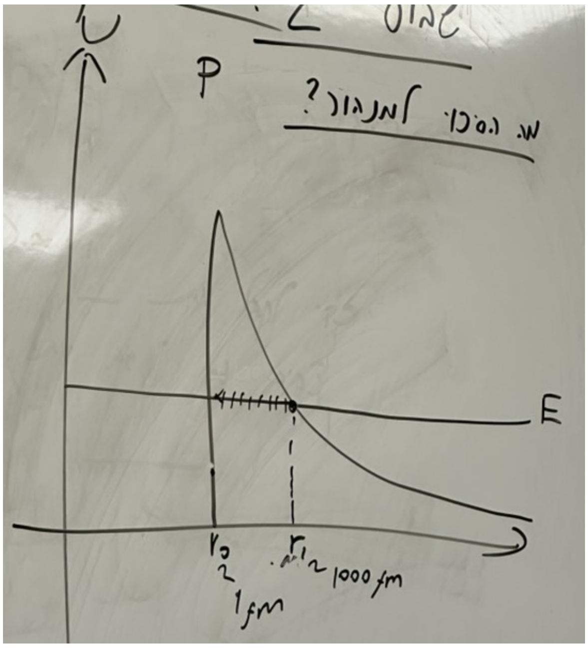
and our potential is

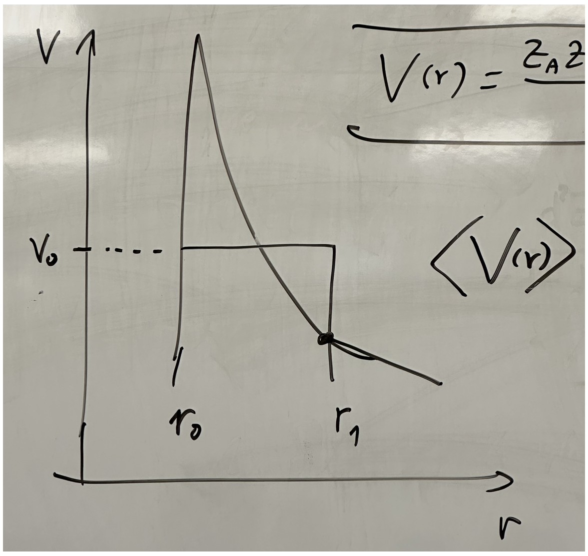
$$V(r) = \begin{cases} \frac{Z_AZ_Be^2}{r} & r > r_0 \\ 0 & r < r_0 \end{cases} \tag{5.19}$$

Solving for this potential is complicated. We'll approximate it using a step function.

Its height should be the average of the potential

$$\langle V(r)\rangle = \frac{\int\limits_{r_0}^{r_1} 4\pi r^2 V(r)\,\mathrm{d}r}{\int\limits_{r_0}^{r_1} 4\pi r^2\,\mathrm{d}r} = \frac{3}{2}E \tag{5.20}$$





and a step function is a problem we know how to solve. Inside the step we have

$$\frac{\hbar^2}{2\mu}\nabla^2\Psi = \left(\frac{3}{2}E - E\right)\Psi \tag{5.21}$$

and since we're working in spherical coordinates

$$\frac{\hbar}{2\mu r}\frac{\partial^2}{\partial r^2}(r\Psi) = \frac{1}{2}E\Psi \tag{5.22}$$

whose solution is

$$\Psi = A\frac{e^{\beta r}}{r} \qquad \beta = \frac{\sqrt{\mu E}}{\hbar} \tag{5.23}$$

and therefore the chance to find a particle in an area δr is proportional to $|\Psi|\,\delta r$ and the probability of tunneling is

$$P_{\text{tunnel}} = \frac{4\pi r_0^2\delta r\,|\Psi(r_0)|^2}{4\pi r_1^2\delta r\,|\Psi(r_1)|^2} = \frac{e^{2\beta r_0}}{e^{2\beta r_1}} = e^{2\beta(r_0-r_1)} \approx e^{-2\beta r_1} \tag{5.24}$$

inserting the parameters we found earlier we find

$$P \approx e^{-\frac{2\sqrt{\mu E}}{\hbar}\frac{Z_AZ_Be^2}{E}} \tag{5.25}$$

while the exact solution is

$$P = e^{-\sqrt{E_G/E}} \tag{5.26}$$

where

$$E_G = (\pi\alpha Z_AZ_B)^2\,2\mu c^2 \qquad \alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137} \tag{5.27}$$

Lecture 6.

Mon, November 17th, 2025

It's worth noting that in the discussion on the probability of tunneling it is more likely the higher the energy, but the cross section for fusion becomes smaller, so it's a tradeoff. Specifically for the pp-chain the numbers are

$$Z_A = Z_B = 1 \qquad \mu = \frac{1}{2}m_p \qquad \mu c^2 = 0.94\,\mathrm{GeV} \tag{6.1}$$

giving us

$$E_G \approx 500\,\mathrm{keV} \tag{6.2}$$

therefore the probability of tunneling is

$$P = e^{-\sqrt{500}} = 2 \times 10^{-10} \tag{6.3}$$

Let's check what this means for the sun. The number of particles in the sun's core is

$$N = 0.1\frac{M_\odot}{m_p} \sim 10^{48} \tag{6.4}$$

So even though the probability of tunneling is low, it definitely happens. To find the rate of the interactions we insert this into 5.9:

$$r_{xi} = \int \sigma n \frac{\mathrm{d}n}{\mathrm{d}v}\,\mathrm{d}v \tag{6.5}$$

$$= \int \sigma \sqrt{\frac{2E}{\mu}} \frac{\mathrm{d}n}{\mathrm{d}E}\,\mathrm{d}E \tag{6.6}$$

$$\tag{6.7}$$

where

$$\frac{\mathrm{d}n}{\mathrm{d}E} = n_i \frac{2}{\sqrt{\pi}} \frac{\sqrt{E}}{(k_B T)^{3/2}} e^{-\frac{E}{k_B T}} \tag{6.8}$$

$$\sigma = \frac{S}{E} e^{-\sqrt{E_G/E}} \tag{6.9}$$

$$S = \frac{\pi \hbar^2}{2\mu} \tag{6.10}$$

giving us

$$r_{xi} = n_i n_x \left(\frac{1}{k_B T}\right)^{3/2} \sqrt{\frac{8}{\pi\mu}} \int e^{-\left(E/K_B T + \sqrt{E_G/E}\right)}\,\mathrm{d}E \tag{6.11}$$

to figure out which particles are most responsible for the fusion process, the high energy ones or the low energy ones, if we designate

$$f(x) = e^{-\left(E/K_B T + \sqrt{E_G/E}\right)} \tag{6.12}$$

then it looks like

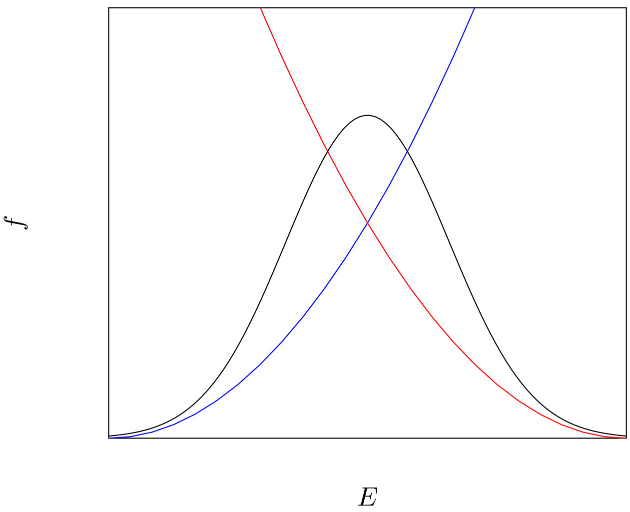


Figure 6.1: The portion of mass of the universe by element in the universe's early times.

and the peak is found at

$$E_0 = \left(\frac{k_B T}{2}\right)^{2/3} E_G^{1/3} \tag{6.13}$$

and specifically for the Sun

$$E_0 = 5\,\mathrm{keV} \tag{6.14}$$

Next we want to solve 6.11. It's not analytically solvable so we approximate it. What's often done in astrophysics is that we approximate things to a power law. Our function looks like this:

$$r_{xi} = An^2 f(T) \tag{6.15}$$

a power law approximation for this would look like

$$r_{xi} \approx An^2 T^\beta \tag{6.16}$$

Why do we do this? Often in astrophysics we look at relationships between values of parameters which span very large ranges, often entire orders of magnitude. for this reason instead of plotting regular linear graphs, we use log-log graphs. If we take a taylor series at some point and look at the first order it's a linear function. A linear line on a log-log graph fits a function of the form T^β. Back to the case of our Sun

$$r_{xi} = An_i n_x f(T) = A\rho_i\rho_x T^\beta \tag{6.17}$$

and if we define

$$x_i = \frac{\rho_i}{\rho} \qquad x_x = \frac{\rho_x}{\rho} \tag{6.18}$$

then

$$r_{xi} = r_0 x_i x_x \left(\frac{\rho}{\rho_0}\right)^2 \left(\frac{T}{T_0}\right)^\beta \tag{6.19}$$

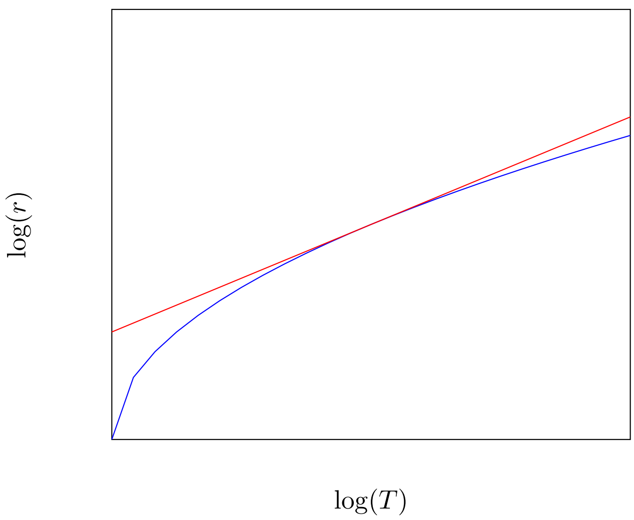


Figure 6.2: The portion of mass of the universe by element in the universe's early times.

where r_0, ρ_o, T_0 are parameters we find from the fit.
The amount of energy produced from all the fusion is

$$\mathrm{d}E = E_0 r_{xi}\,\mathrm{d}t\,\mathrm{d}V \tag{6.20}$$

and inserting what we found so far

$$\mathrm{d}E = E_0 r_0 x_i x_x \left(\frac{\rho}{\rho_0}\right)^2 \left(\frac{T}{T_0}\right)^\beta \mathrm{d}V\,\mathrm{d}t \tag{6.21}$$

which we are able to rewrite to an integral over the mass by using $\mathrm{d}m = \rho\,\mathrm{d}V$:

$$\mathrm{d}E = \left(\frac{E_0 r_)}{\rho_0}\right) x_i x_x \left(\frac{\rho}{\rho_0}\right) \left(\frac{T}{T_0}\right)^\beta \mathrm{d}m\,\mathrm{d}t \tag{6.22}$$

where we can define

$$\epsilon_0 = \frac{E_0 r_0}{\rho_0} \tag{6.23}$$

which has units of energy produced per gram material per second, giving us

$$\frac{\mathrm{d}^2 E}{\mathrm{d}m^2} t = \epsilon_0 x_i x_x \frac{\rho}{\rho_0} \left(\frac{T}{T_0}\right)^\beta \tag{6.24}$$

and if we insert $\beta = 4$ which fits our Sun and

$$\epsilon_0 = 0.1\,\mathrm{erg\,g^{-1}\,s^{-1}} \qquad \rho_0 = \mathrm{g/cm^3} \qquad T_0 = 10^7\,\mathrm{K} \qquad x_i = x_x = 0.7 \tag{6.25}$$

then we find

$$\epsilon_\odot = 7.4\,\mathrm{erg\,g^{-1}\,s^{-1}} \tag{6.26}$$

then the luminosity produced is

$$\mathrm{d}L = \epsilon\,\mathrm{d}m = \epsilon \cdot 4\pi r^2\,\mathrm{d}r\rho \tag{6.27}$$

which gives us

$$\frac{\mathrm{d}L}{\mathrm{d}r} = \epsilon \cdot 4\pi r^2 \rho \tag{6.28}$$

2.7 Diffusion of Photons

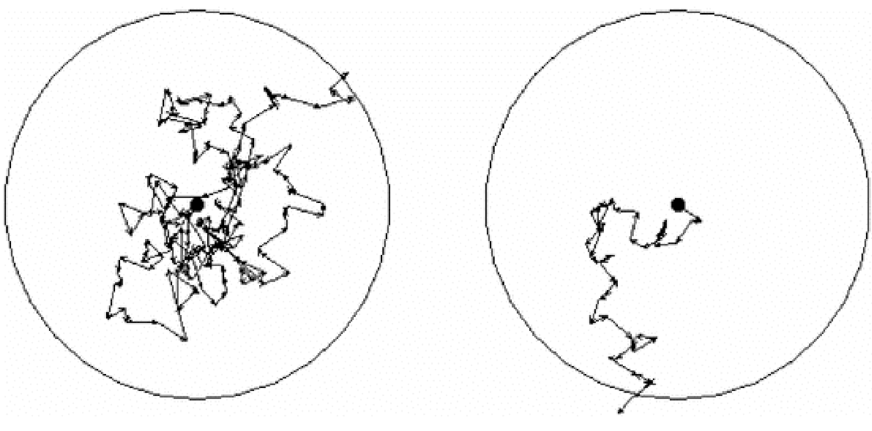


Figure 6.3: Example of Compton scattering Brownian motion (Random walk) of a photon.

Suppose we have a star and inside a photon is created in the middle and moves around inside until it reaches the outside. This process is called a random walk. Each step is of length λ and the star's radius is R.

$$R^2 = |\mathbf{R}|^2 = \left|\sum \mathbf{r}_i\right|^2 \tag{6.29}$$
$$= \sum_i \sum_j \mathbf{r}_i \cdot \mathbf{r}_j \tag{6.30}$$
$$= \sum_{i=1}^{N} |r_i|^2 + \sum_{i \neq j} \mathbf{r}_i \cdot \mathbf{r}_j \tag{6.31}$$
$$= N\lambda^2 \tag{6.32}$$

therefore the number of steps until the photon escapes is

$$N = \left(\frac{R}{\lambda}\right)^2 \tag{6.33}$$

some typical numbers are

$$R = R_\odot \qquad\qquad \lambda \approx 0.1\,\mathrm{cm} \tag{6.34}$$

which gives us

$$N \approx 5 \times 10^{23} \tag{6.35}$$

if we want the distance the photon will travel in total then

$$d = \lambda \cdot N \approx 4 \times 10^{22}\,\mathrm{cm} = 16\,\mathrm{kpc} \tag{6.36}$$

and the time it takes the photon to travel this distance is

$$t = d/c \approx 50\,000\,\mathrm{yr} \tag{6.37}$$

If want a more accurate description of the process, we can use the diffusion equation. We won't develop it from scratch but begin from Fick's Diffusion Law (A development from scratch can be found in Feynman's lectures volume 1 chapter 42):

$$\mathbf{J} == -D\nabla n \tag{6.38}$$

where J is the flux of particles, $D = \frac{\lambda v}{3}$ is the diffusion parameter. We'll use this formula for photons. The free traversal is given by $\lambda = \frac{1}{\sigma n} = \frac{1}{\kappa \rho}$ where σ is the action cross section and $\kappa = \frac{\sigma}{m}$ is the opacity. So for photons we'll perform the substitutions

$$v \to c \qquad D \to \frac{\lambda c}{3} = \frac{c}{3\kappa\rho} \qquad \rho \to u_{rad} = aT^4 \qquad J \to f_{rad} = \frac{L_r}{4\pi r^2} \tag{6.39}$$

which gives us

$$\frac{L_r}{4\pi r^2} = f_{rad} = -\frac{1}{3}\frac{c}{\kappa\rho}\frac{\mathrm{d}}{\mathrm{d}r} aT^4 \tag{6.40}$$

rearranging differentiating we find

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{L_r}{4\pi r^2} \tag{6.41}$$

In conclusion we have four differential equations:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -G\frac{M_r\rho}{r^2} \qquad \frac{\mathrm{d}M_r}{\mathrm{d}r} = 4\pi r^2\rho \tag{6.42}$$

$$\frac{\mathrm{d}L_r}{\mathrm{d}r} = 4\pi r^2\rho\epsilon \qquad \frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{L_r}{4\pi r^2} \tag{6.43}$$

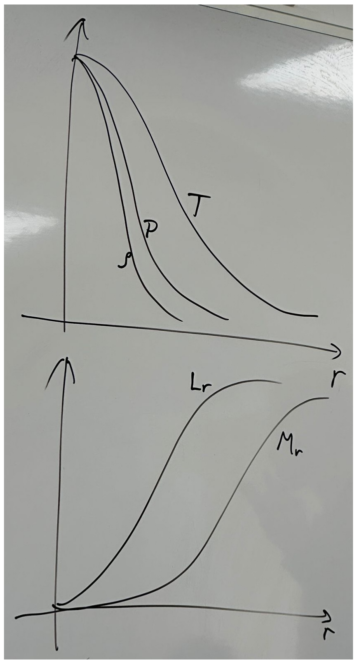
together with the state equation P, the energy production equation ϵ, and the opacity κ we have all the tools we need to describe the structure of stars.

Lecture 7.

Thu, November 20th, 2025

For an approximate set of solutions we can assume that

$$T, P, \rho|_{r=R_*} = 0 \qquad M_r = L_r|_{r=0} = 0 \qquad \begin{cases} M_r|_{R_*} = M \\ L_r|_{R_*} = L \end{cases} \tag{7.1}$$



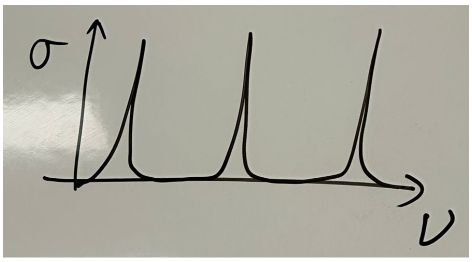
In the process of diffusion, photons can undergo several kinds of interactions

Bound-bound (Excitation) $X + \nu \to X^*$.

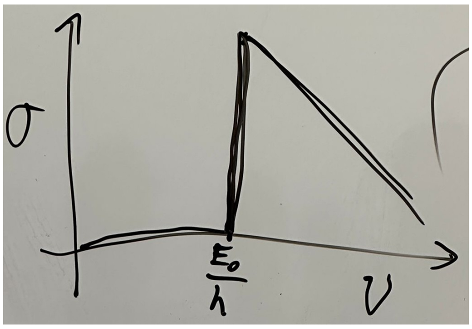
Bound-free (Ionization) $X^n + \nu \to X^{n+1} + e$.
With $\sigma = \sigma_0 \left(\frac{h\nu}{E_0}\right)^{-3}$ for hydrogen and atoms with one electons.

Free-free absorption (Reverse bremsstrahlung)

Free-free e-scattering (Thomson) which has a well known action cross section $\sigma_T = \frac{8}{3}\pi r_e^2$ where $r_e = \frac{e^2}{m_e c^2}$. Additionally, if there are ions in the area then if $\lambda \ll r_p$ there can happen Compton scattering and if $\lambda \gg r_p$there can happen Rayleigh scattering.



(a) Bound-bound (excitation)



(b) Bound-free (Ionization)

Lecture 8.

Mon, November 24th, 2025

3 L-M ties

Looking at the HR diagram we see that there appears to be a connection between mass and luminosity. We'll dwell on this for a while. This also connects to its life span and evolution. Observationally, if we plot the luminosity as a function of the mass, we find something that looks like this

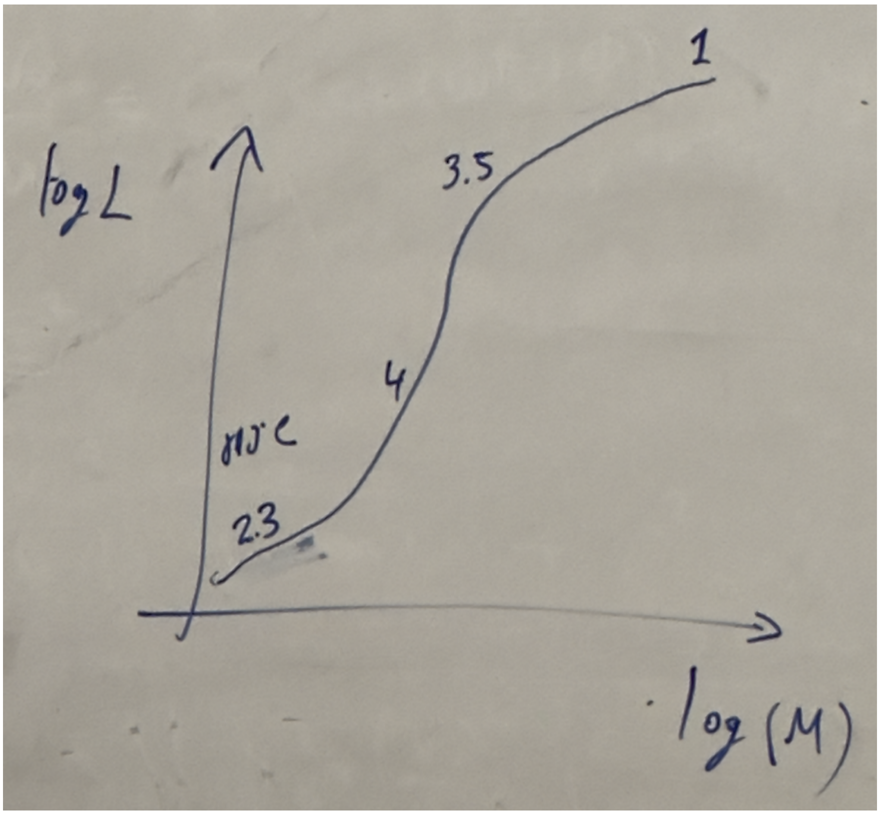


Figure 8.1: A diagram of the L-M relation

and using the relation

$$\frac{L}{L_\odot} = A\left(\frac{M}{M_\odot}\right)^\alpha \tag{8.1}$$

we can write this as a table To get a feeling for why this behaves like this, we'll do the following.

$M/M_\odot$	A	α
< 0.43	0.23	2.3
0.43-2	1	4
2-55	1.4	3.5
> 55	3.2×10^4	1

We'll assume

$$\kappa = \kappa_{\text{Thomson}} = \text{const.} \tag{8.2}$$

which is a good approximation for massive stars (2 to 55 solar masses). We'll also assume a power law of the form

$$M_r \propto r^\alpha \qquad P \propto r^\beta \qquad T \propto T^r \tag{8.3}$$

therefore the stellar structure equations become

$$\beta \frac{P}{r} \propto \frac{M\rho}{r^2} \qquad \alpha \frac{M}{r} \propto r^2 \rho \qquad \gamma \frac{T}{r} \propto \frac{L\kappa\rho}{r^2 T^3} \tag{8.4}$$

rearranging we find

$$P \propto \frac{M\rho}{r} \qquad M \propto r^3 \rho \qquad L \propto \frac{T^4 r}{\rho} \qquad P \propto \rho T \tag{8.5}$$

Inserting D into A we find

$$T \propto \frac{M}{r} \tag{8.6}$$

and inserting that into C

$$L \propto \frac{(M/r)^4 r}{\rho} = \frac{M^4/r^3}{\rho} \propto M^3 \tag{8.7}$$

For extremely massive stars ($> 55M_\odot$), the radiation pressure is dominant, at which point D no longer holds and instead

$$P \propto T^4 \tag{8.8}$$

which inserted into A gives

$$T^4 \propto \frac{M\rho}{r} \tag{8.9}$$

and inserting that into C

$$L \propto \frac{(M\rho/r)\, r}{\rho} = M \tag{8.10}$$

3.1 The Most Massive Stars

Let's look at the outer layer of a very massive star. Using the radiation propagation equation

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{L_r}{4\pi r^2 c} \tag{8.11}$$

we can rearrange it to

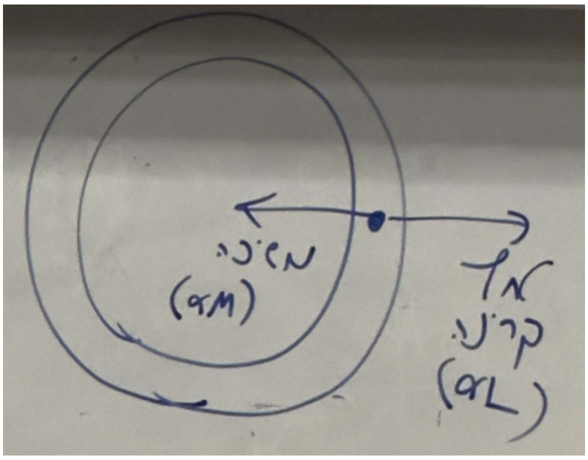
$$\frac{1}{3}\frac{\mathrm{d}}{\mathrm{d}r}\left(aT^4\right) = -\kappa\rho\frac{L_r}{4\pi r^2 c} \tag{8.12}$$

and remembering that $\frac{1}{3}aT^4$ is the radiation pressure

$$\frac{\mathrm{d}P_{\mathrm{rad}}}{\mathrm{d}r} = -\kappa\rho\frac{L_r}{4\pi r^2 c} \tag{8.13}$$

on the other hand, from the hydrostatic equations we have

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM_r\rho}{r^2 c} \tag{8.14}$$



setting our two expression to equal we find

$$\frac{GM_r\rho}{r^2} = \kappa\rho\frac{L_r}{4\pi r^2 c} \tag{8.15}$$

isolating L_r and inserting $r = R$

$$L_r = 4\pi c\frac{GM}{\kappa} \tag{8.16}$$

this expression for luminosity is often called Eddington Luminosity. This expression gives us the maximum luminosity a star can have before it becomes unstable. If for a given star's parameters the observed luminosity is higher than the Eddington luminosity, then the radiation pressure will begin to overpower its hydrostatic pressure which will destabilize the star. Let's plug some numbers in to get a feel. Inserting the constants we have

$$L = 3.8 \times 10^4 \frac{M}{M_\odot} L_\odot \tag{8.17}$$

and for a mass of $M = 60 M_\odot$

$$L = 2.3 \times 10^6 \, \mathrm{L}_\odot \tag{8.18}$$

3.2 The Lifespan Of Stars

For the Sun we can write

$$t_\odot \simeq \frac{\eta M_{\text{core}} c^2}{L_\odot} \sim 10 \, \text{Gyr} \tag{8.19}$$

If we look at stars with different masses then

$$t_* \propto \frac{M}{L} \propto M^{-3} \implies t_* = t_\odot \left(\frac{M}{M_\odot} \right)^{-3} \tag{8.20}$$

3.3 Convection

Convection becomes important if

$$t_{\text{rad}} \gg t_{\text{dyn}} = \frac{l}{c_s} \tag{8.21}$$

A more concrete criteria for convection

Let's look at a element of gas with density ρ and temperature T at radius r in a star, which moves to $r + dr$ and its parameters change to $\rho + \delta\rho$ and $T + \delta T$. The criterion for convection is

$$\rho_{\text{element}} + \delta\rho_{\text{element}} < \rho_{\text{env}} + d\rho_{\text{env}} \tag{8.22}$$

or

$$\delta\rho < d\rho \tag{8.23}$$

then the element will float upwards. For a more accurate derivation we'l assume this is an adiabatic process, which lets us say

$$P \propto \rho^\gamma \qquad\qquad \frac{\delta\rho}{\rho} = \frac{1}{\gamma}\frac{\delta P}{P} = \frac{1}{\gamma}\frac{dP}{P} \tag{8.24}$$

we'll also assume there is a pressure equilibrium with the star. Therefore

$$P = \frac{\rho}{m} \kappa T \tag{8.25}$$

which gives us

$$\frac{d\rho}{\rho} = \frac{dP}{P} - \frac{dT}{T} \tag{8.26}$$

We can now insert both 8.24 and 8.26 into 8.23 we find

$$\frac{1}{\gamma}\frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}r} < \frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}r} - \frac{1}{T}\frac{\mathrm{d}T}{\mathrm{d}r} \tag{8.27}$$

rearranging

$$\frac{\mathrm{d}T}{\mathrm{d}r} < \frac{\gamma - 1}{\gamma}\frac{T}{P}\frac{\mathrm{d}P}{\mathrm{d}r} \tag{8.28}$$

both the differentials are smaller than 0, so we'll multiply both sides by -1 and find

$$\left|\frac{\mathrm{d}T}{\mathrm{d}r}\right| > \frac{\gamma - 1}{\gamma}\frac{T}{P}\left|\frac{\mathrm{d}P}{\mathrm{d}r}\right| \tag{8.29}$$

The effect this has is that it improves the transfer of heat which in turn balances it out recursively, and after a while the greater symbol will become equal

$$\left|\frac{\mathrm{d}T}{\mathrm{d}r}\right| = \frac{\gamma - 1}{\gamma}\frac{T}{P}\left|\frac{\mathrm{d}P}{\mathrm{d}r}\right| \tag{8.30}$$

which is a stable state. If 8.29 is satisfied, we replace the radiation diffusion equation for the structure of stars with the convection equation 8.30.

Lecture 9.

Thu, November 27th, 2025

I missed this lecture.

Lecture 10.

Mon, December 1st, 2025

I missed this lecture.

Lecture 11.

Thu, December 4th, 2025

I missed this lecture.

Lecture 12.

Mon, December 8th, 2025

Last week we started talking about the beginning of the collapse of the core of a massive star. During the cire's collapse two processes which cause instability take place:

Destruction of Irons $\gamma + Fe \to 13He + 4n$ which consumes 124 MeV and $\gamma + He \to 2p + n$ which consumes 28 MeV.

Combination of protons and electrons $e+p \to n+\nu_e$ which is possible when $k_B T \gg (m_n - m_p)\, c^2$

The energy consumed in the first process is roughly

$$N_{Fe} \approx \frac{M_{ch}}{m_{Fe}} = \frac{1.4M_\odot}{56m_p} \sim 3 \times 10^{55} \implies E_{loss} = N_{Fe} \cdot \frac{(124 + 28.13)\text{MeV}}{500\,\text{MeV}} \sim 2 \times 10^{52}\,\text{erg} \tag{12.1}$$

After both processes our star's material is rich in neutrons, which are fermions, and therefore Pauli's exclusion principle applies to them too. Everything is the same as for elections, the only difference being that they have a different λ_{DB}. A similar analysis to electrons can be done with two differences: using the neutron mass instead of the electron's, and setting $Z/A = 1$. This results in

$$R_{ns} = 2.3 \times 10^9 \frac{m_e}{m_n} \left(\frac{M}{M_\odot}\right)^{-1/3} = 11\,\text{km} \left(\frac{M}{M_\odot}\right)^{-1/3} \tag{12.2}$$

which is about 500 times smaller than a white dwarf. The density in comparison to a whie dwarf will be

$$\frac{\rho_{ns}}{\rho_{wd}} \sim 500^3 \sim 10^8 \implies \rho_{ns} \sim 10^{14}\,\text{g/cm}^3 \tag{12.3}$$

If we look at the density of material in the nucleon of an atom it is

$$\rho_{nuc} = \frac{m_p}{\frac{4\pi}{3}\,(1\,\text{fm})^3} \sim 4 \times 10^{14}\,\text{g/cm}^3 \tag{12.4}$$

So the density of a neutron star is in comparable to that of the nucleus of an atom. You could think of the entire thing as a single atom nucleus with atomic number 10^{57}. Recalling the expression for $M_c h$:

$$M_{ch} = 0.21 \left(\frac{Z}{A}\right)^2 \alpha_G^{-3/2} m_p \tag{12.5}$$

setting the relation we had for neutron stars

$$M_{ns} = 0.21\alpha_G^{-3/2} m_p = 5.6M_\odot \tag{12.6}$$

which is the upper limit for the mass of a neutron star. A neutron star more massive than that won't hold itself together and collapse further. Observations indeed haven't found any neutron stars heavier than (or even close to) that mass. More accurate calculations give a result significantly lower. The reason for this is that general relativistic effects become important at this scale. Relativistic effects become relevant when

$$\frac{E_{Gr}}{mc^2} \sim \frac{\frac{GM_{ns}m}{R_{ns}}}{mc^2} = \frac{GM_{ns}}{R_{ns}c^2} \sim 0.2 \tag{12.7}$$

Another important effect we haven't mentioned yet is the strong nuclear force. Since the neutron star effectivelty behaves like a giant nucleus, the strong force will start to have an effect. We don't quite understand what its effect is though. If we want to look at how much energy is expected to be released in the collapse of a core into a neutron star

$$\Delta E_{gr} = \left| \frac{\frac{3}{5}GM^2}{R_0} - \frac{\frac{3}{5}GM^2}{R_{ns}} \right| \approx \frac{\frac{3}{5}GM^2}{R_{ns}} \tag{12.8}$$

and if we insert typical quantities for neutron stars

$$M = 1.3 M_\odot \qquad\qquad R_{ns} = 11\,\mathrm{km} \tag{12.9}$$

we find

$$\Delta E_{gr} \sim 3 \times 10^{53}\,\mathrm{erg} \tag{12.10}$$

This energy has to go somewhere and this causes a supernova. This is the leading process which causes supernovas. This kind of supernova is called a core-collapse supernova. The luminosity we see from this process is

and the total energy we observe is

$$E_\gamma = \int \mathrm{d}t L = 3 \times 10^{49}\,\mathrm{erg} \tag{12.11}$$

which is only 0.01% of the available energy. Where does the rest of the energy go? Maybe it is ejected from the supernova. The ejected mass is about $(1-10)\,M_\odot$ and observations show that its velocity is $v \sim 10^4\,\mathrm{km\,s^{-1}}$. This gives us an energy of

$$E_k \sim \frac{1}{2} M_{ej} v^2 = 3 \times 10^{51}\,\mathrm{erg} \tag{12.12}$$

which is 1% of the total energy, so there must be another explanation. Approximately 99% is ejected as neutrinos. The density in the center of the supernova is extremely high and the temperature is very high, which means there are very many photons whizzing around close to each other, and some of them collide with each other which results in

$$\gamma + \gamma \to e + e^+ \to \begin{cases} \nu_e + \overline{\nu}_e \\ \nu_\mu + \overline{\nu}_\mu \\ \nu_\tau + \overline{\nu}_\tau \end{cases} \tag{12.13}$$

The conditions are so extreme that sometimes even the neutrinos go through collision processes. The reason this process doesn't happen in the sun is that there just isn't enough energy there. The rest energy of the electron is $m_e c^2 = 0.5\,\mathrm{MeV}$ which photon collisions can't reach in the sun.

Other types of explosions: Type 1a Supernovas. There's a white dward just chilling on its own but a red giant approaches it and the white dwarf starts sucking material out of the red giant. It then eventually reaches M_{ch} and explodes.

In regular stars there is a negative feedback loop where fusion causes temperature to increase, which increases the pressure, which increases the radius of the star, which lowers the temperature, which lower the rate of fusion. In contrast in a white dwarf fusion causes an increase in temperature which doesn't affect the pressure, so it increases the rate of fusion again, and so on. This is called a type 1a supernova.

Another kind of explosion is a gamma ray burst. The energy released from GRBs is $E \sim 10^{51}$ erg which is all released over about 0.1 seconds to a minute. GRBs are an active research area and it's theorized that there are several types.

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